

```
create database college;
```

Explain: Creates a new database named **college**.

```
use college;
```

Explain: Ask the permission to perform.

```
create table students (  
    rollno int primary key,  
    name varchar(20),  
    age int,  
    department varchar(20)  
);
```

Explain: Creates a table named students with columns like rollno , name , age ,department.

```
alter table students  
add city varchar(20);
```

Explain: Add a new column named city to store the student's city.

```
alter table students  
change department dept varchar(20);
```

Explain: Rename the column department as dept.

```
insert into students  
values  
(101,'Anu',20,'CSE','Chennai'),  
    (102,'Karan',20,'IT','Madurai'),  
    (103,'Rohan',20,'CSE','Coimbatore'),
```

```
(104,'Siva',20,'CSE','Tirunelveli'),  
(105,'Priya',20,'ECE','Chennai'),  
(106,'Radha',20,'ECE','Madurai'),  
(107,'Jack',20,'IT','Salem');
```

Explain: Inserts multiple students records into student's table.

```
select * from students;
```

Explain : Display all records from the students table.

```
select name,dept from students;
```

Explain: Display the name and department of all students.

```
update students  
set city = 'Trichy'  
where rollno = 105;
```

Explain: Update the city of the student where rollno is 105 to Trichy.

```
update students  
set age = 22  
where rollno = 102;
```

Explain: Update the age of the student where rollno is 102 to 22.

```
delete from students  
where rollno = 107;
```

Explain: Delete the student whose rollno is 107.

```
select * from students  
where dept = 'CSE';
```

Explain: Display all records who belong to the CSE department.

```
SELECT COUNT(name)
FROM students
WHERE dept = 'CSE';
```

Explain: Count the number of students in CSE department.

```
select * from students
order by rollno desc;
```

Explain: Display all students records sorted by rollno.

```
select dept
from students
group by dept;
```

Explain: Display the unique department names by grouping.

```
select count(*)
from students
where dept = 'IT';
```

Explain: Count the total number of students in the IT department.

```
select dept,count(*)
from students
group by dept
having count(*)<3;
```

Explain: Displays the departments that have less than 3 students.

Insert Table:

	rollno	name	age	dept	city
▶	101	Anu	20	CSE	Chennai
	102	Karan	22	IT	Madurai
	103	Rohan	20	CSE	Coimbatore
	104	Siva	20	CSE	Tirunelveli
	105	Priya	21	ECE	Trichy
	106	Radha	22	ECE	Madurai
	106	Radha	20	ECE	Madurai
	107	Jack	20	IT	Salem
*	NULL	NULL	NULL	NULL	NULL

Name and department table:

	name	dept
▶	Anu	CSE
	Karan	IT
	Rohan	CSE
	Siva	CSE
	Priya	ECE
	Radha	ECE
	Radha	ECE
	Jack	IT

Update and delete table:

	rollno	name	age	dept	city
▶	101	Anu	20	CSE	Chennai
	102	Karan	22	IT	Madurai
	103	Rohan	20	CSE	Coimbatore
	104	Siva	20	CSE	Tirunelveli
	105	Priya	20	ECE	Trichy
	106	Radha	20	ECE	Madurai
	NULL	NULL	NULL	NULL	NULL

Students table by dept name:

	rollno	name	age	dept	city
▶	101	Anu	20	CSE	Chennai
	103	Rohan	20	CSE	Coimbatore
	104	Siva	20	CSE	Tirunelveli
*	NULL	NULL	NULL	NULL	NULL

Count table:

	COUNT(name)
▶	3

	count(*)
▶	1

Order by rollno:

	rollno	name	age	dept	city
▶	106	Radha	20	ECE	Madurai
	105	Priya	20	ECE	Trichy
	104	Siva	20	CSE	Tirunelveli
	103	Rohan	20	CSE	Coimbatore
	102	Karan	22	IT	Madurai
	101	Anu	20	CSE	Chennai

Dept name by group by table:

	dept
▶	CSE
	IT
	ECE

Having table:

	dept	count(*)
▶	IT	1
	ECE	2